

## 5.8 A panorama of regression models to come

The linear model of this chapter makes three kinds of claim: about the *mean* of the response (a linear combination of known columns), about its *variability* (a common variance, no correlation) and, in the normal version, about its *distribution*. Every model in the rest of the book relaxes one or more of these claims, or keeps them and asks harder questions about them. This section surveys the route.

### 5.8.1 Three components of a regression model

It helps to see any regression model as three separate choices.

1. **A distribution for the response given the regressors:** normal, binomial, Poisson, gamma, or something less specified, such as “any distribution with this mean and variance”.
2. **A predictor** that combines the regressors: the linear predictor  $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , or a sum of smooth functions, or a sum that includes random terms.
3. **A link** between the predictor and the mean,  $g(E(Y_i)) = \eta_i$ , where  $g$  is a known monotone function.

The normal linear model chooses the normal distribution with constant variance, the linear predictor, and the identity link  $g(\mu) = \mu$ . Changing one choice at a time generates the models below.

### 5.8.2 The ladder

| Model                                 | What changes                                                                                                                                              | Where          |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Linear model, second moments          | the starting point: geometry, optimality, estimability, computation                                                                                       | Chapters 5–10  |
| Normal linear model                   | exact tests, intervals, multiple comparisons, designed experiments                                                                                        | Chapters 11–18 |
| Departures from the assumptions       | diagnostics, transformations, resampling, measurement error, causation                                                                                    | Chapters 19–25 |
| Many regressors                       | collinearity, shrinkage, $p > n$ , selection, boosting                                                                                                    | Chapters 26–30 |
| General covariance and random effects | $\text{Cov}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{V}$ ;<br>$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\epsilon}$ | Chapters 31–33 |
| Generalized linear models             | non-normal distribution and a link                                                                                                                        | Chapters 34–41 |
| Nonparametric and additive models     | smooth functions in the predictor                                                                                                                         | Chapters 42–44 |
| Distributional regression             | regressors act on more than the mean                                                                                                                      | Chapter 45     |

**The normal linear model and its designs.** With normal errors, the least squares estimator has an exact normal distribution and the residual sum of squares an exact chi-squared distribution (Theorem 7.5.1). This gives  $t$  and  $F$  tests and exact confidence regions (Chapters 11 and 12), and procedures for many comparisons at once (Chapter 13). Analysis of variance and designed experiments (Chapters 15–18) are linear models whose columns are indicator variables, often not of full rank; their theory rests on Chapter 6 and Chapter 8.

**When the assumptions fail, or there are many regressors.** Part V asks what happens when the mean is misspecified, the variance is not constant, the errors are correlated or not normal, the regressors are measured with error, or the question is causal; some remedies, such as robust standard errors based on Eq. 5.5.3 (Chapter 21), stay inside the linear model. Part VI treats large  $p$  and nearly dependent columns, where ridge regression, principal components and the lasso trade a little bias for less variance. Regression trees, random forests and neural networks also estimate  $m$ , judged by prediction error rather than by the meaning of coefficients; the book does not develop them, but Chapters 29 and 30 compare predictors and treat boosting.

**General covariance and mixed models.** The general Gauss–Markov model replaces  $\sigma^2\mathbf{I}$  by  $\sigma^2\mathbf{V}$  (Chapter 31), and generalized least squares is least squares in a different inner product (Section 6.9). Linear mixed models write the response as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\varepsilon},$$

where  $\mathbf{b}$  is a vector of random effects shared by related cases, such as repeated measurements on one person or pupils in one school (Chapters 32–33). The RAND data of Example (Self-rated health), with the same people observed in several years, are of this kind.

**Generalized linear models.** For a binary response the mean is a probability, and a linear predictor can leave  $[0, 1]$ . For a count the mean must be positive and the variance grows with the mean. Generalized linear models keep the linear predictor, choose a distribution from an exponential family, and connect the two by a link: the logit  $\log\{\mu/(1-\mu)\}$  for binary responses (Chapter 35), the logarithm for counts (Chapter 37). Coefficients act additively on the scale of the link, so in a Poisson model with log link  $e^{\beta_j}$  is the factor by which the mean changes per unit of  $x_j$ . Chapters 34–41 develop the theory, overdispersion, correlated responses and missing data.

**Nonparametric and additive models.** Polynomials, as in Example (A curved trend and a seasonal cycle), force a global shape on a curve. Splines and smoothers let the data choose the shape locally (Chapters 42–43), and additive models replace each term  $\beta_j x_j$  by an unknown smooth function  $f_j(x_j)$  (Chapter 44). Much of this is still penalized least squares on a model matrix of basis functions.

**Beyond the mean.** Quantile regression models a chosen quantile of  $Y$ , such as the median or the tenth percentile, as a linear function of the regressors, and so describes how the whole conditional distribution shifts. Distributional regression lets the regressors act on the location, the scale and the shape of the response distribution through separate predictors, turning process (b) of Figure 5.1.1 into a model rather than a violation (Chapter 45).

**The Bayesian thread.** A prior distribution on  $(\boldsymbol{\beta}, \sigma^2)$  turns the likelihood into a posterior. For the normal linear model with a conjugate prior the posterior is available in closed form (Theorem 7.6.2), and it reappears for intervals (Theorem 12.6.2) and for generalized linear models (Chapter 39).

### 5.8.3 A first look beyond the linear model

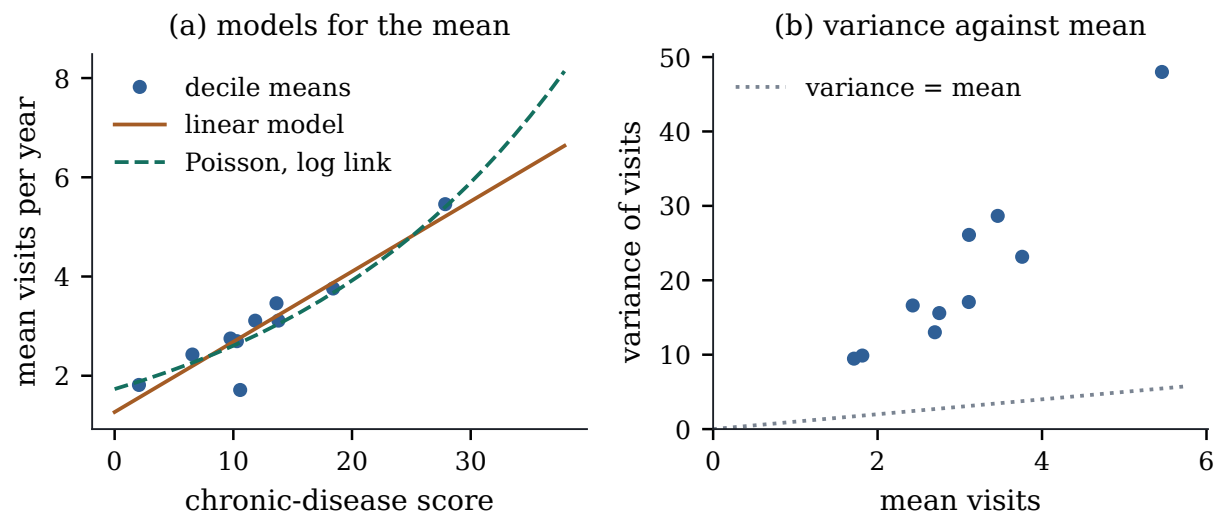


Figure 5.8.1: Annual physician visits in the RAND Health Insurance Experiment, grouped into ten groups by a chronic-disease score. (a) Group means with a linear model and a Poisson regression with log link. (b) Group variances against group means; the dotted line is the Poisson relation  $\text{variance} = \text{mean}$ .

### Example. (Physician visits are counts)

The number of physician visits in the RAND data is a count. It is zero for 31.2 percent of person-years, and its variance is 7.09 times its mean. Group the person-years into ten groups by the chronic-disease score, and compute the mean and variance of visits in each (Figure 5.8.1). The variance rises with the mean, and in every group it is between 4.8 and 8.8 times the mean. (L2) fails badly.

Two models for the mean are fitted to the disease score  $x$ . The linear model gives  $1.268 + 0.1416x$  visits. The Poisson regression with log link (Chapter 37) gives  $\exp(0.549 + 0.0408x)$ : each unit of the index multiplies the mean by  $e^{0.0408}$ , 4.17 percent more visits. Over the bulk of the data the two curves are close, and both follow the group means. They differ in shape at high values of the index, where the log-linear mean grows ever faster and the linear one at a constant rate, and where the data are too sparse to decide between them. They differ more in what they assume about variability. The linear model assumes a constant variance; the Poisson model assumes variance equal to the mean. The data contradict both. Chapter 38 shows how to keep the log-linear mean and let the variance be a multiple of it.

The Poisson fit is not computed by ordinary least squares, but its estimating equations have the same form as the normal equations:  $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$ , where  $\hat{\boldsymbol{\mu}}$  is the vector of fitted means. The residuals are again orthogonal to every column of  $\mathbf{X}$ . The listing checks this.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
x = rand["disea"].to_numpy()
X = sm.add_constant(x)
```

```

bins = pd.qcut(x, 10, duplicates="drop")          # ten groups by disease score
groups = pd.DataFrame({"y": y, "x": x}).groupby(bins, observed=True)
table = groups.agg(x=("x", "mean"), mean=("y", "mean"), var=("y", "var"))
print(table.round(2).to_string(index=False))
print(f"share of zeros {np.mean(y == 0):.3f}; variance/mean overall {y.var() / y.
      mean():.2f}")

linear = sm.OLS(y, X).fit()
poisson = sm.GLM(y, X, family=sm.families.Poisson()).fit()
print("linear model:      ", np.round(linear.params, 4))
print("Poisson, log link:", np.round(poisson.params, 4))
mu = poisson.fittedvalues
print("X^T (y - mu_hat) =", X.T @ (y - mu))

```

[Run this code online](#)

## 5.8.4 Exercises

### 5.8.4.1 A. Check your understanding

#### Exercise A1.

For each response, name the rung of the ladder you would start from, and say which of the three components (distribution, predictor, link) departs from the normal linear model: (a) whether a loan is repaid; (b) blood pressure measured monthly on the same patients; (c) the number of insurance claims per policy; (d) the 90th percentile of birth weight as a function of the mother's age; (e) crop yield in a randomized block experiment.

### 5.8.4.2 B. Practice

#### Exercise B1.

Let  $Y_i$  be binary with  $\Pr(Y_i = 1) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  (the *linear probability model*). Show that (L1) holds but (L2) fails, with  $\text{Var}(Y_i) = \mu_i(1 - \mu_i)$ . Show that least squares is still unbiased, find its covariance matrix from Proposition 5.5.7(b), and explain why fitted values outside  $[0, 1]$  are possible.

*Solution.*  $E(Y_i) = \Pr(Y_i = 1) = \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , which is (L1).  $\text{Var}(Y_i) = \mu_i - \mu_i^2$ , which varies with  $i$  unless all the  $\mu_i$  are equal or symmetric about  $1/2$ , so (L2) fails. Unbiasedness needs only (L1) (Theorem 5.5.1(a)). With independent cases,  $\boldsymbol{\Sigma} = \text{diag}(\mu_i(1 - \mu_i))$  and Eq. 5.5.3 gives  $\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$ . Nothing in least squares constrains  $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$  to  $[0, 1]$ : the estimate is a linear function of  $\mathbf{y}$ , and at design points far from the centre of the data the fitted line can cross 0 or 1 even when every true probability is inside. The logit link of Chapter 35 removes the problem.

#### Exercise B2.

For independent  $Y_i \sim \text{Poisson}(\mu_i)$  with  $\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , write the log-likelihood and show that its gradient is  $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu})$ . Deduce that when  $\mathbf{X}$  contains an intercept the fitted

means sum to  $\sum_i y_i$ , and that for the intercept-only model the fitted mean is  $\bar{y}$ .

### 5.8.4.3 C. Going deeper

#### Exercise C1.

In process (b) of Example (Same regression function, different models) the error standard deviation is  $0.15 + 0.25x$ . Suppose instead  $\log \text{Var}(Y_i) = \gamma_0 + \gamma_1 x_i$  with unknown  $\gamma$ . Propose a two-step procedure: fit the mean by least squares, then regress  $\log \hat{\varepsilon}_i^2$  on  $x_i$ . Show that, when the errors are normal and  $n \rightarrow \infty$  with  $p$  fixed, the slope of the second regression estimates  $\gamma_1$  consistently, while its intercept estimates  $\gamma_0 + E(\log \chi_1^2)$ . (Argue first with the errors  $\varepsilon_i$  in place of the residuals, then explain why the difference does not matter in the limit.) Compute this bias,  $-\gamma_E - \log 2 \approx -1.27$ , where  $\gamma_E$  is Euler's constant. (Chapter 45 treats models of this kind by likelihood.)